

Institute of Actuaries of India

Subject SA1 – Health and Care Insurance

March 2017 Examination

INDICATIVE SOLUTION

Solution 1:**i) Proposed Product Design**

This product is aimed at incentivizing policyholders who are either healthy or intend to be healthy along with a desire to proactively managing their lifestyle. The product aims to offer discount in premiums based on the active management of health and it could lead to positive selection by healthy lives or those who believe that they will continue to manage a healthy lifestyle.

The overall morbidity experience of these lives are expected to be light and hence a reduction in premium will act as an incentive for them to continue to renew the policy. Hence it could help in reducing anti-selective lapses by healthy lives.

Further, it may also motivate them to continue to proactively manage health in order to get discounts in premium.

It is important to note that the discount in premium is available irrespective of the claim history. If the individual's health parameters that are being tracked meet the requirements, he/she would be eligible for the discount. This may be appreciated by the policyholders who may not feel dissuaded in making a claim. This is different from No Claim Bonus feature that rewards the policyholder for not making a claim.

This product will also increase the level of engagement with the insured which may help the company to cross sell or upsell. This product will enable the insurer to help the policyholder in managing a healthy lifestyle.

However, the extent of discount in premium is likely to be a key to the success of this product. This is because the product will require individuals to share their personal data at regular intervals (which is otherwise not required in existing products) with the insurer which may be a concern to some individuals. Further there may be a concern of security of the personal data.

So, the premium differentials have to be attractive to motivate the customer to buy this product. Specially, for younger lives who tend to be healthier and are otherwise not interested in buying health insurance.

Having said that for younger, the proportion of claims due to chronic / lifestyle related illnesses could be low and hence maintaining physical fitness may not greatly impact the overall morbidity experience. For older lives though, the reduction in premium could be high due to better health outcomes that could lead to better morbidity experience (specially due to chronic /lifestyle illnesses)

Another challenge is to continue the motivation levels throughout the policy year to manage health in lieu of reduction in premium. This is because the insurer would want to ensure that the policyholder is active throughout the year and not just at the time of renewals. So maintaining the level of motivation to remain physically active through the year could be a challenge.

For some customers, it may not be comfortable to wear the device all time and may miss out on discount even if he is healthy.

It could be a challenge to ensure that the device is worn only by the insured and no one else. The system has to be developed in a manner that it can identify and warn the insurer if the device is worn by any person other than the insured.

If the company decides not to offer the wearable device then the individual has to have a wearable device to be eligible to buy this product. It could pose a challenge given that wearable devices are a relatively new technology in the market and are expensive. The insurer would have to find a way to make the

devices available at a reasonable rate in order to make the product a success. Partnering with device manufacturers would have to pass through regulatory approvals.

If the device were to be offered with the policy then it could become a very capital intensive as the cost of the devices will have to be funded by the insurer upfront. Further, if the cost is recouped over few policy years then it may become sensitive to lapses. Insurer may find it difficult to recover the device from a lapsed policy.

Having said that if a similar product does not currently exist in the market then it may help the insurer in increasing their revenue and volume. Also, until other insurers launch a similar product, there will be no competition on the price and hence the insurer can price with a higher margin to recoup the cost of infrastructure and uncertainty regarding the ultimate morbidity experience.

The cost of developing this product could be substantial if the company doesn't have the required technology infrastructure. Hence it is important to establish the need for such a product.

The marketing team could undertake a survey of the customers as well as distributors to understand customer sentiments. They could also look for anecdotal evidence by analyzing the sales pattern of wearable devices, sports equipment or increase / decrease in gym membership to understand whether there is an increased health consciousness. Further, the company can also look at other markets where this product concept has been tried and whether it has been a success.

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ii) Pricing Approach

The pricing is very crucial for the success of this product as mentioned above. The pricing approach should explain how the base premium is calculated and then the discounts offered. The premium discount should reflect the expected morbidity experience.

The risk premium calculation could be similar to the risk premium of the existing indemnity product. The burning cost should be calculated based on insurer's own experience. Since the insurer has been in existence for a long time, the availability and credibility of that data should be sufficient to derive the burning cost. This should then be adjusted for medical inflation and expected morbidity experience of the target population for the proposed product.

Further, as per the proposed design, the premium remains unchanged at renewal if the policyholder hasn't actively managed his health. The premium reduction is only offered to lives that meet the specified criteria.

Those who become eligible for discount could be assumed to be healthy lives. Hence, the residual category (i.e. the category that doesn't qualify for any discount) as a result is likely to have relatively less healthy lives. The experience from this category is likely to be poorer than aggregate experience and hence the base premium for the residual category is likely to be higher.

The overarching assumption is that the aggregate experience of the portfolio will remain unchanged. Those who get discount will pay lower premium which will be offset by those who will pay higher premium for potentially poorer experience.

One approach to derive the premium rates for this category is to first derive the premium at the portfolio level. Then derive the levels of premium discount and proportion of policies that will be eligible for such discount. The premium for the residual category will be calculated such that the overall experience of the portfolio is 100% of the experience of the aggregate portfolio.

For better risk categories, where premium discount has to be offered, the pricing actuary has to establish which parameters are reasonable predictors of the morbidity experience and the extent of impact on expected morbidity experience.

Further the frequency of data captured to by the insurer has to be decided.

Data for some of the parameters e.g. number of steps, sleep pattern, BMI, heart rate etc. can be made available by the wearable device. These data points then have to be converted into objective fitness measures.

For example: following aspects could be tracked:

1. Number of steps
 2. Sleep pattern
 3. Heart Beat
 4. Activities like running, cycling,
 5. For other activities like yoga, swimming and physical work-out, it can be based on user inputs.
- However this could lead to false declarations

The correlation between these parameters and their likely impact on the morbidity experience has to be established.

Another aspect will be to decide whether blood test or medicals will be carried out periodically (or at the time of inception and / or renewals) to gather other vitals which are useful in establishing the health status of the individual. This is only possible if the health checkup is funded by the insurer and is mandatory to be eligible to discount. The cost of checkup will also need to be built. The feasibility of this option needs to be decided. If not then only the parameters that can be obtained by the wearable device needs to be taken for pricing.

The impact of regular physical activities on acute and chronic illnesses could vary which needs to be considered. The actuary may want to analyze the pattern of claims by age and gender into acute and chronic illnesses. If physical fitness helps in reducing or delaying onset of chronic illnesses, then the discount offered could be higher in those age bands where proportion of claims due to chronic / lifestyle illnesses are high.

Further for each risk factor, a range of values will need to be decided that will segregate lives into different buckets. E.g. for Body Mass Index (BMI) a possible range of values would have to be identified which can then be used to segregate lives into different risk classes. A person with low BMI would need to be classified under low risk class and vice versa. Similar classification will be required for other factors.

Then, the combination of these risk factors and their likely impact on health outcomes will then need to be quantified. Depending on the likely saving in claims cost for each of the risk class, the discount in premium needs to be calculated. The number of discount levels will depend on the availability of granular level at the time of pricing.

The premium structure needs to be simple in order to be understood by potential applicants or distributors.

Given that the concept is new, the insurer is unlikely to have granular level data for the all the parameters and their impact on the claims experience from its own data. The insurer may wish to approach the reinsurer to get data related to fitness levels and associated experience.

The insurer may also approach established device manufacturers who may have detailed data for a long time.

The actuary also needs to decide if the effect of physical activity is same across all ages. That is whether the people who are healthy today despite being physically active will start to get sick over time as they get older. This could mean that the proportion of people in higher discount category may reduce over time as the portfolio starts to age. Hence the portfolio mix assumption becomes very important.

Further, if more people qualify for premium reduction than priced then and the claim experience remains as expected then premiums at the portfolio level may become insufficient. The actuary needs to test the sensitivity of the portfolio mix on the profitability of the product.

Since this is a yearly renewable product, the risk of mispricing is reduced.

The expense assumption will be very crucial. Whether the wearable device is given as part of the product or the customer will have to buy his own. If it's the former then the cost of device has to be factored into pricing. If the cost is to be recouped over few policy years then the renewal rate assumptions becomes crucial. Lapse sensitivity needs to be tested.

As the cost of each device is likely to be substantial and it will increase the initial capital requirements. The cost of the capital could increase the premium which may reduce the attractiveness of the product. It will also make the product sensitive to lapses.

If the policyholders are supposed to bring their own device then compatibility could be an issue. The insurer will have to deal with multiple device manufacturers and their method of capturing data. This will increase cost at insurer's end to structure them for its own use.

The other pricing assumptions could be as per the current pricing basis for indemnity products.

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iii) a) Underwriting and Sales

The product is likely to attract people who are healthy and hence there is likely to be some positive selection. It is assumed that the product requires individuals to own the device. Hence, it can be reasonably assumed that the socio economic profile of the lives opting for this product is likely to be high as it will have those individuals who not only understand the importance of a wearable device and but can also afford it .

In general, the underwriting practices could be very similar to the existing indemnity product. A moratorium based underwriting approach could help in ease of issuance.

However, as the premium reduction is based on certain rating factors, it's important to get the initial baseline estimates at the inception.

Depending on the rating factors, a medical test may be necessary. However, conducting medical tests may not be feasible for each and every policy as it will increase the cost of issuance as well as delay the issuance process. Hence rating factors have to be obtained in the proposal form as well as the initial reading from the wearable device without the need of a medical test.

Medical tests could be conducted for older age or higher sum assured or as per the existing underwriting policy of the company

The data recording per each application will be more than the existing policy applications. Hence the underwriting systems have to be updated and personnel need to be trained.

Sales

If a similar product is not currently available in the market then this product is likely to attract attention. The insurer may want to capture the market by aggressive advertising. The advertising could be focused towards the target market of a technologically aware population who is educated and relatively wealthy.

The marketing cost, as a result, will increase which could be compensated by potentially increased volume.

The focus of the marketing will also be to increase awareness towards benefits of proactive health management. The insurer has to be very particular in its marketing message of offering premium discount which is different than the No Claim Bonus scheme which could be perceived by some as way to dissuade customers in making a claim.

Since there is a pre-requisite that the customer should have a wearable device at the time of buying the policy, it is likely that the target customer base for this product is likely to be well educated and aware of the latest technologies. Moreover it is possible that most of these customers belong to Metro or Tier 2 cities.

The product is more complicated than the existing products and hence will require significant training. The insurer will have to decide the channels that will be used.

The direct-to-customer channel could be used as the insurer can directly engage with the customer who is already aware of the benefits of good health. The social media platforms can be used to generate leads and then insurer can reach out to the potential individual.

Agents and own sales force could also be trained and equipped to sell the policies.

The insurer can use other avenues to generate leads e.g. gyms /fitness clubs, fitness consultants, adventure sports clubs, wearable manufacturers or online platforms subject to regulatory framework allowing for such lead generation.

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b) Administration and other considerations

The administration of this product will be very complicated. The insurer may be required to upgrade its IT infrastructure.

Each policy sold will need to interact with the insurer's server so that it can receive data from the wearable devices. This can make the IT systems vulnerable to cyber-attacks. The insurer will have to ensure that the policyholder data is well protected and hence has to develop additional data security measures.

Further, depending on the frequency of the data submitted, the insurer is likely to receive significant data points for each individual. This data needs to be stored securely and for a long duration. The increased storage cost will need to be factored in. The number of parameters that will be used to offer discount will also need to be captured in the policy admin system.

The staff strength will need to be augmented and need to be trained in dealing customer queries and complaints towards hardware and software issues which could be new to the insurer. The insurer may need to outsource some of the functions keeping in mind the regulatory restrictions.

Other considerations

One of the main considerations could be regulatory approval. The product, being very complicated, will need to be discussed thoroughly with the regulator. The regulator needs to be convinced that the premium rates are adequate and that the cross subsidies are kept to the minimum. Given the uncertainties involved, the insurer could use the “Pilot product” provision in the recent health regulations for seeking regulatory approvals.

Another consideration will be availability and cost of wearable devices. These devices till now are mainly available in major cities and are costly. It is important that the sale of the device increase in order for insurer to increase the potential size of the market for insurance. Further, insurer’s offering has to be compatible with these devices. The insurer could think of having a partnership with one or two major device manufacturers with a significant geographical presence and then market the offerings together. This, again, has to be within the regulatory framework.

Further, most of these devices operate via smartphones where by the manufacturer develops an application that is installed on customer’s smart phone. This application is then synced with the device and data is transmitted to the phone from the device. The insurance company’s servers should be equipped to get the data from multiple sources.

One crucial aspect will be internet connectivity. For the data to be transmitted, the internet connectivity and strength of connection is important. In Metro cities, internet connectivity may not be a challenge but the speed could be. Hence the insurance company needs to test whether it can receive data even at lower bandwidths / internet speed.

Finally, the success of the product is heavily dependent on the fact that the policyholder has to continuously be motivated to follow a certain health routine. Otherwise if the consistency and discipline is not followed, the insurer may find it difficult to extend the premium discount.

(4)**iv) a) Mispricing Risk**

The mispricing risk arises when the actual experience is better or worse than the expected experience and the quantum of the difference is rather large. The experience variance could be mainly due to the following reasons:

1. Poor selection or anti-selection whereby the lives opted for insurance is different than what was assumed in pricing.
2. The true underlying experience is very different than expected

For the latter, if the data that was used to derive the assumptions was incorrect or insufficient, it could lead to unpredictable experience.

As the wearable devices are fairly new concept and in the stage of evolution, it is unlikely that credible data is available with anyone.

For this product, there is a risk that the parameters that will be used to offer premium discount may not be the true predictor of the morbidity experience.

E.g. if the insurers uses number of steps per day as one of the predictors for better morbidity experience and offers discount in premium if someone walks for say 10,000 steps per day on average. However, it could be that the actual claim experience is independent of this variable as number of walking mainly helps in managing lifestyle or chronic illness. However the insured is still exposed to infectious or other

acute illnesses and hence the actual experience may suggest that number of steps may not be a good predictor for predicting morbidity experience.

Following actions could be considered for mitigating mispricing risk

1. Reinsurance: the reinsurer can provide help in pricing, product design, technical knowhow as well as sharing of the risk.
2. Allowing for margins for adverse deviation. However, allowing for a higher margin could make the offering uncompetitive
3. Frequent review of experience: this will help in checking the actual experience against expected
4. Offering reviewable premiums – it may be difficult to review premiums especially in the first 3 years of launch.
5. Sensitivity testing

(3)

b) Reputational Risk

The insurer has established itself over the last 10 year and now has built a reputation that it needs to protect.

The reputational risk under this product arises from the following scenarios:

1. The product fails to meet customer's expectations: the product promises to offer reduction in premium for healthy lives. If the mechanism used by the insurer is stringent such that only few policyholders become eligible for the discount then it may not be appreciated by those who believe that they are healthy but didn't qualify for the discount. This may lead to bad publicity. As the underlying population is likely to be educated and technologically aware, their presence on the social network platform could be significant. This could lead to bad publicity spreading quicker than expected and impacting the brand image of the insurer.

For mitigation, the insurer will have to ensure that the product design and its limitations are well explained to the policyholders. Further insurer has to put an active monitoring mechanism in place to ensure that activity levels are actively monitored and being communicated to the customer at regular intervals.

2. The third parties that are integral to the product offerings don't deliver as expected. E.g. some of the wearable devices malfunction or don't submit the required data to the insurer on time. The insurer will be unable to offer premium reduction if the data is not submitted correctly and as per the required format. The customer however is not at fault but wouldn't appreciate if he /she doesn't get discount even if he/she has been physically active.

For mitigation, the insurer will have to put mechanism in place where by faults in data capturing is caught and rectified at the earliest.

3. There is a data breach where by the personal data stored by the insurer gets leaked or hacked. The security mechanisms have to be regularly tested and increased measures should be taken to ensure that any data breach is contained at the earliest

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c) Counterparty Risk

The major counterparty risk under this product arises if the insurer decides to use external device manufacturers to obtain data from the insured lives. Any changes in their business model could directly impact the insurer's business.

For e.g. if one of the device manufacturers decides to close its business or declares itself bankrupt and hence would no longer be able to provide support either to the insurer or its own customers then it would adversely impact insurance company. Unless the insured find a replacement of the device, he/she won't be able to submit data to the insurer.

To mitigate this risk the company may have an exclusive agreement with few device manufacturers in the early years after the product launch. If the concept proves successful then it could either acquire any of the manufacturers or starts manufacturing its own devices.

Another major counterparty risk under this product will arise if the insurer outsources the IT infrastructure development and maintenance. The insurer is unlikely to have in-house expertise to develop say the application or systems that could interact with the wearable devices. In such a scenario, it will probably need to outsource the development work. If the counterparty fails to deliver then it may impact insurer's ability to deliver its promises. To mitigate such risk, the insurer should gradually plan to move these processes in-house so that its dependency on external party reduces as the insurer starts to gain experience.

Similarly if the insurer engages with an analytical company to co-develop the product and rely on their output to price the business. To mitigate, the insurer should aim to build its analytical capabilities over time. Until then, the insurer should have a mutually beneficial agreement in place that ensure that the interests of both the parties are protected.

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v) a) Manufacture the device

If the insurer decides to manufacture the devices then the counterparty risk is minimised. The hardware would be compatible with the software and vice versa.

It will be easier for the insurer to develop an integrated system rather than develop a system keeping in mind compatibility with multiple device manufacturers.

This will also help the insurer to be sure of the quality of the materials used in the hardware and there will be consistency in the data recorded.

The insurer can also provide after sales service for its own device. A prompt service will be well appreciated by the policyholders and may increase the persistency.

Further, it will be easier to make changes in product design or add / remove risk parameters. Future upgrades of both the hardware and the software is easier to manage with less dependency on external providers

Having said, it would require a significant capital outlay to manufacture the device. Moreover the insurer may not have the competency or its core strength to design the device and then develop a product concept around it. Even if it does, it could take significant amount of time and money.

Further, if the product doesn't sell in expected volumes then the investment cost may not be recovered.

(3)

b) Partner with existing device manufacturers

This would significantly reduce the capital outlay

The insurer can bring the product into the market in a short span to time.

The device manufacturers are the specialist and hence insurer can benefit from their expertise.

The manufacturers will have the infrastructure to provide after sales service.

Insurer can also get data from their existing user base which it could use for pricing as well as cross selling

If the insurer deals with few selected manufacturers then it could develop its software that is compatible to these devices

However, there is a counterparty risk as explained earlier

The association may require insurer to pay the manufacturers in order to receive data.

(3)

vi) Proposal Evaluation

Offering as a rider to existing indemnity policyholders:

This would help in reducing anti-selective lapses whereby the healthy lives port to the new product to pay a lower premium while the unhealthy lives continue to stay in the existing product. The proposal is to offer the benefit as a rider whereby the policyholder, at the time of renewal, will have the option to take the rider. Once the rider has been opted, the vitals of the policyholder will be tracked over the year and depending on the health status, a discount in premium could be offered at the time of the next renewal.

There is a potential self-selection here whereby policyholder who opt for the rider are likely to be those who believe that they are healthy and hence can benefit from it to get reduction in premium. The unhealthy policyholder may not be interested.

This could impact the existing product's profitability if it has a small proportion of lives with very poor morbidity experience which is currently being funded by a large proportion of healthy policyholders. Hence, it is important to price the rider premium in such a manner that the profitability of the product remains unchanged if a significant proportion of healthy lives opt for the rider.

Since the current product has been in existence for a long time, it is possible that the portfolio has started to age or the initial underwriting effect may have worn off. This means more unhealthy lives in the existing portfolio that may have completed all the waiting periods. This could also impact the take up rate of the rider.

Offering as a rider to existing CI policyholder:

A typical CI product offers a lump sum benefit on first ever diagnosis or undergoing treatment of the listed Critical Illness conditions. The product is usually bought to cover for the financial loss in the event of diagnosis of CI. Hence the SA under the policy is a lump sum and is representative of potential loss of income and expense incurred in treating the illnesses. Typically, benefit is paid only once during the policy lifetime. Once a claim is paid the policy ends.

Further, the illnesses that usually get covered are the ones that have high severity but very low frequency. Many of the covered illnesses are caused due to non-lifestyle related factors. These illnesses may or may not be directly linked to the levels of physical activities. For e.g. Cancer is one of the covered illnesses and some types of cancers (sites) does have any association with the physical fitness and could happen due to other reasons. Then there are illnesses e.g. heart attack where physical fitness could prevent or delay the onset of cardiovascular diseases.

Typically, the CI fixed benefit products have a level premium. Hence offering reduction at the end of each year mayn't be feasible as level premiums don't correctly represent the risk premium for a given insurance period as the cost of insurance is spread over the policy term to arrive at the level premium.

Hence, the actuary needs to judge whether the rider can be correctly priced depending on the illnesses covered and whether the resultant premium offers a strong value proposition to the customer.

(7)

vii) Approach to calculate rider premium

The rider premium should be such that total rider premium should be approximately equal to the reduction in premium offered.

This is based on the assumption that the overall morbidity experience of the portfolio remains unchanged. This is because premium reduction is irrespective of the claims experience.

So any loss of premium due to offering the premium discount should be recouped from the rider premium.

An assumption regarding the take up rate is required. Another assumption will be to set a proportion of lives (that opted for the rider) that will get premium reduction.

For e.g. let's assume 10% is the take up rate for a portfolio of 1,000 lives. Of which 30% of the members are likely to qualify for say a 10% reduction in the indemnity premium.

If the average indemnity premium is Rs. 5,000 for a given age and Sum Insured then the total reduction in premium will be $30 * 500 = 15,000$.

This will be shared by 100 (10% of 1,000) policyholders i.e. Rs. 150. This means that by paying Rs 150, the member is likely to get Rs 500 discount in premium at the time of renewal.

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[50 Marks]

Solution 2:

i) Five key points to be considered in choosing a local partner for joint venture

In order for the joint venture to work successfully, both the partners need to be fully aligned. Hence, consensus on various aspects like Investment horizon, management control, product philosophy, market positioning, long term strategic ambitions etc. are crucial in making the joint venture work.

Financial aspects like the ability to infuse required capital break even, profitability, return on Capital etc. Flexibility on shareholding pattern is also very important considerations in choosing the joint venture partner. This will ensure financial alignment and resource availability as per needs.

The company is a large health insurance company in the overseas market. Hence, it will have deep technical expertise in the Health Insurance space – pricing, reserving etc. It will hence, look for complementing skills in the joint venture partner: Someone who understands the local market is close to customers and understands their needs and preferences, existing relationship with regulators etc.

Good brand reputation: Trust factor with the customers, regulator etc. This will be important as the local customers are more likely to be able to relate with the local partner at-least initially. Also, Insurance is a business of trust. Hence, products and services that the local company is currently offering will also be an influencing factor.

The current regulations in India do not allow one company to be significant shareholder in 2 or more companies. Hence, the partner should not be the shareholder of an existing Insurance Company or should be willing to sell the current state before entering the new JV.

(10)

ii) a) Key changes in the proposed guidelines

The Health Insurance Regulations, 2013 were replaced by IRDAI (Health Insurance) Regulations, 2016 which were notified on 18th July, 2016. In terms of various provisions of the said Regulation, IRDAI issued Guidelines on product filing procedures for products relating to Health Insurance Business for compliance by all Insurers and TPAs, as may be applicable.

The Product Filing Guidelines set out the procedure and the requisite forms for filing various types of health insurance products, such as, new health insurance products, Pilot Products, Health Package Products and Non-Life Package Products under the File & Use Procedure.

The provision for submitting a lawyer's certificate while filing such products with the IRDAI has now been omitted.

The Product Filing Guidelines also mandates setting up of a "Product Management Committee" (PMC) by General Insurers and Health Insurers. The PMC is, among other functions, required to carry out a due diligence process and record its concurrence on various product related risks for all product filings under File & Use before any application for a product is sent to the IRDAI for approval.

A significant change brought about by the Product Filing Guidelines is the introduction of the Use & File procedure for group health insurance products. As per this procedure, a group product may be launched without the prior approval of the IRDAI subject to the following stipulations:

- The product must be approved by the PMC;
- The Insurer shall obtain a Unique Identification Number for every insurance product from the IRDAI by duly informing them of the name of the product proposed to be launched; and
- The product shall be filed with the IRDAI within seven days from the date of approval by the PMC in the prescribed form.

Life Insurers are not permitted to file group health insurance products under the Use & File Procedure and are required to follow the procedure of File & Use as prescribed under the Product Filing Guidelines for these products.

In addition, the Product Filing Guidelines provide detailed guidance on Wellness Features and Benefits that an Insurer may offer in a health insurance product. The Product Filing Guidelines also provide an extensive procedure for the withdrawal of products.

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b) Challenges and Benefits

While the Health Insurance regulations covered the provisions, the guidelines provided a clear framework for product filings.

This gave a clear framework to the Insurance companies thereby avoiding any non-compliance with the underlying regulations. The guidelines thus might increase compliance cost initially but is expected to bring efficiency in a longer run.

The dropping of the lawyer's certificate requirement for product filing is a welcome change.

Pilot products were only talked about first in the Health Insurance regulations 2016. The guidelines thereby provide further details and clarifications on the filing for such products.

The requirement to set up a product management committee by all insurers and products to be first approved by the PMC puts more onus/responsibility to the insurers. For Group health insurance, product filing to follow use & file makes the role of the PMC very critical.

Regarding Product withdrawals, the insurers would now be required to follow a clear set of guidelines.

The guidelines formalize the wellness features/benefits provided by insurers. While it puts certain restrictions (no payments to third parties towards wellness features etc., it clearly lays down the rules, thereby removing ambiguity and increasing policyholder confidence.

The guideline also requires a customer information sheet to be filled. This shall provide product details to the customers thereby ensuring that the customers understand the product well. This should lead to lower miss-selling, higher renewals and hence, lower expenses per policy in a longer run. This is also expected to reduce claim rejections and overall lower claim expenses.

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iii) Potential Implications of CI Product withdrawal

Adverse media reaction may cause negative customer sentiment. If the reaction is severe enough there may be brand damage, possibly leading to lower volumes sold of the company's other products. There may also be an additional administrative burden from dealing with worried existing policyholders.

The book will run off with a consequent reduction in the volume in-force. This will make the statistical analysis of experience less and less viable and administration costs per policy will gradually increase. There will come a point when the administration becomes economically unviable. The insurer would need to consider selling off the residual business at some point.

Existing customers may be more likely to lapse their products which will further reduce the value of the book. Sick customers, or customers in poor health, are unlikely to lapse, so there will be a selective lapsing effect worsening the morbidity experience of the remaining book and therefore reviewable premiums may need to increase, with corresponding implications.

There will be greater volatility of experience on the shrinking run-off portfolio which may lead to needing higher reserves and capital requirements for that business. Capital requirements may also increase due to the reduced level of diversification now that a whole product line is no longer sold.

Claims management may become less robust as the portfolio reduces, increasing the risk of acceptance of claims that might otherwise have been rejected (e.g. due to exclusions).

The ability to maintain sales volumes is a powerful motivator to maintain the claims service standards – after closure to new business, distributors are likely to be cynical as to the future of the in-force book. Hence distributors may rebroke their clients to a competitor product.

As the CI product is being sold at a loss, this will stop more loss-making business being put on the books. This will enable the company to use its capital in a more efficient manner e.g. to fund the new business strain on more profitable lines of business and to focus its attention and expertise in more appropriate directions.

There may be reinsurance implications and certain treaty clauses may get triggered.

There may be tax implication depending on how significant this product range has been within the portfolio of business.

The insurer needs to decide whether it will introduce a replacement new product range. If a new product is launched there may be selective lapse and re-entry. If there is no replacement product range, then the insurer will likely need to reduce its staff levels with a potential impact on morale which will incur an extra “one-off” cost through redundancy payments.

Other impacts on customers:

Existing customers should find that the policy continues to be serviced and claims met, at least for the medium term (until the volume of in-force policies falls to below a critical level – as noted above); to this extent they should not be adversely affected.

Depending on the cut-off point used, pipeline customers may or may not find that their policy is still taken on the books.

If policies contain contractual rights to options to increase the cover or term, these options must still remain available to them.

The company is reasonably successful. If the critical illness product is significant in the market, this could cause a market shift – e.g. increasing prices for new products.

The insurer will need to communicate with customers and other stakeholders. Customers may become aware of media activity or otherwise become concerned about the future of their contract; they may not be clear about whether their contract remains in-force.

Regulator’s involvement

The company will have to ensure that the regulator’s requirements are met with respect to product withdrawal. The provisions put forward in the Guidelines for product filing will have to be met.

(8)

iv) Experience of Micro Insurance IP v/s. Standard IP

The main differences are likely to be the claim inception rates and the claim termination rates. The proposed product will likely have higher claim inception rates due to less stringent definitions of incapacity, less stringent underwriting, the more manual nature of the policyholder’s normal occupation and a higher incidence of pandemics due to sanitation arrangements.

Differences in claim termination rates are less clear cut. Lower levels of medical facilities may lead to longer claims than expected for comparable incapacities as will the less stringent definitions of incapacity, the more manual nature of the policyholder’s normal occupation, the potentially higher levels of initial serious disability (e.g. landmines, mining accidents etc) and the differing levels/effectiveness of the claims management system. Also some diseases e.g. AIDS, leprosy may not lead to recovery. However higher levels of mortality may terminate claims earlier than expected.

The speed of return to work may depend on state of the economy/level of employment; there may also be a different cultural attitude towards returning to work.

Claim rates may be higher due to higher levels of fraud. Large volumes of small policies make fraud more difficult to spot and the cost of procedures to identify fraud may outweigh savings. The local culture may

make investigation difficult e.g. reluctance to talk to investigators. Also, in a culture where insurance is not commonplace, there may be less appreciation of the consequences of a fraudulent claim.

Policy documents may have to be translated in local languages. Poorly translated policy documentation may lead to a higher level of disagreements. There may be potential for bad publicity due to refused claims.

It may be difficult or costly to travel to a doctor in remote locations to confirm sickness. The quality/professionalism/training of doctors will also have an effect.

There is a greater potential for anti-selection and moral hazard. There may be more non-disclosure (e.g. relating to PECs) e.g. due to lack of understanding of the importance of full disclosure.

Withdrawal rates will probably be different. They may be higher due to pressures on income which may be very variable, e.g. weather dependant, drought etc or due to lower appreciation of the value of the policy or they may be lower due to higher recognised need. Withdrawal rates may also depend on level of competition.

Expenses are likely to be higher. Initial expenses will be higher due to the need to recover the costs of developing a new product and training the staff locally for the new product. New systems may be required to cope with the large volumes of data. The cost of new systems might be amortized over a few years.

Per policy expenses will be higher. Individual premiums will be small so relative costs of banking and accounting will be increased.

Premium payment frequency may be monthly not annually and payments may be in cash mostly. There may be record keeping issues due to the large numbers of transactions. Collectors may not pass on premiums or they may be stolen in transit. Potential for disputes – policyholders may not keep receipts etc.

It may take some time for the experience to be built and stabilized. Also, data capturing may be difficult. All this may lead to uncertainty in experience in a short run and hence a requirement to hold higher reserves.

Potentially suitable matching assets are unlikely to be available

Volumes of business may be higher depending on the potential size of the target market. There will be less diversification of business by target market, as this is mainly the very poor population who will be insured, or by geographic location.

Availability or reinsurance may also affect the experience. Reinsurers may or may not want to underwrite this business or may be very selective.

The challenge in improving experience in this micro-insurance product via claims management and underwriting is to keep the costs proportionate to the small size of the policies.

Reduce higher claims experience by becoming involved in community projects e.g. provision of clean water, or in providing preventative health measures e.g. vaccination, or providing education on the overall benefits of healthy lifestyles. Could aim to work closely with the government on educational initiatives.

Amend the product design, for example limit benefit payment period, have less generous claims definitions and appropriately worded exclusions, more initial underwriting, stricter claims management, provide rehabilitation support.

Use of reinsurance, if available

Use of tight and clearly worded terms and conditions, e.g. to reduce potential for non-disclosure.

Reduce higher expenses by entering into joint venture with distribution partner or outsourcing some functions e.g. distribution, policy administration, claim handling, negotiate fixed per policy servicing cost, train more (cheaper) local staff.

Invest in good IT infrastructure for efficient payment and record keeping.

Use collective investment vehicles. As well as reducing investment costs and the cost of expertise, this may also limit adverse performance relative to other insurance companies. The potential for adverse investment performance can also be reduced by increasing the level of asset/liability matching, if possible.

Reduce poor withdrawal experience by offering renewal discounts and proactive premium collection and monitoring/reminders. Have appropriate commission structure depending on distribution channel.

Reduce fraud by educating policyholders about the effects of fraud on the availability and cost of insurance, careful selection and remuneration of local doctors and setting up clinics with medical staff directly employed by the insurance company (may also reduce claim experience if involved in preventative measures). The insurer may need to join with other insurance providers to share the costs of these initiatives.

Use domestic medical expertise for periodic audit of quality standards.

Reduce the effects of concentration of business by finding a joint venture partner or co-insurer to help diversify.

(13)

v) Conditions to manage the benefit provided and possible risk mitigation for reinstatement benefit

The reinstated Sum Insured should be utilized only for claims for new illness or disease for the insured person during the same policy year and any claims for any illness suffered by other members in case of family floater

The reinstatement benefit should be triggered only post exhausting your existing coverage including the no claims bonus (sum Insured increases as a result of no claims made in the previous years)

Only Basic Sum Insured may be reinstated, not the Sum Insured increases as a result of any claim bonuses

The reinstated benefit should not be carried forward to the next year

The reinstated sum Insured can be used for only future claims made by the Insured Person. Hence, in case the basic sum insured plus no claim bonus is not adequate for the first claim itself during a policy year, the reinstated benefit may not trigger at that moment

The Restore Sum Insured should only be applied once for the Insured Person or family floater during a Policy Year

(4)

vi) Cost of providing the Sum Insured reinstatement benefit

The pricing of the Sum Insured reinstatement benefit very much depends on the conditions that govern this benefits. Here, we are assuming that this benefit is governed by the conditions mentioned in the answer to previous section.

Broadly the price of this benefit can be determined as per the following formula:

The probability of claim in a policy X Probability of claim amount exceeding SI if claim is made X Probability of another claim in same policy year for another illness X The average claim amount in second instance = Risk premium of SI reinstatement

On the face of it this may sound simple but has lot of nuances, like:

- 1) There can be multiple episodes of claims in both first and second instances, therefore, the probability needs to take care of this
- 2) Similar to above the average claim size should be sum total of all the claims in both the instances
- 3) The accumulated bonus will further complicate this as it will be at different levels for different SI's
- 4) The illness for which claims has been made before exhaustion of base SI and accumulated bonus will not be payable from reinstated SI
- 5) For family floater the above condition will be even more complex as the same person cannot claim for same illness but other members can be allowed to claim for illness for which members previously have claimed before SI exhaustion

Considering all these complexities it is recommended to use Monte Carlo Simulation to price this benefit.

In order to price this, the first step is to arrange the data in the following manner:

- a) Exposure should be computed in terms of number of lives for Individual SI and both number of lives and number of policies for the Family Floater plans
- b) Exposure should be divided by age, gender, etc.
- c) Claims should be arranged by all the variables, like, age, gender, etc. This should be done by per illness and for each episode of claims separately. This means that the no. of claims in a policy should not be aggregated rather they should be counted separately each time a claim is made for any illness
- d) Metrics like incidence rate and claim amount per exposure should be determined
- e) Claim distribution should be determined for frequency as well as severity (for example lognormal and negative binomial)

Once above is in place, the model should be built year wise.

In the year one the claim count and claim amount should be simulated for relevant age band and SI combination. The count and amount should be aggregated. Whenever aggregate claim amount exceeds given SI, the reinstatement of SI should be triggered. If there are still claims in that policy basis the simulation, the next illness of claims should be checked against the prior claims and if it is different the

amount of such claim should be included in determining the risk premium of SI reinstatement for that year.

Then we should move to next year. The SI for next year will be same as base SI if there was claim in this policy otherwise it should include 'No Claim Bonus' also. The same process as above is to be repeated to find out cost of reinstatement for second year. The claim amount would go up by medical inflation.

The same process should be repeated for no. of years considering claims inflation and average risk premium for the period should be taken for which the price will remain unchanged.

For Floater Plans we can use the claims count, average claim amount and type of illness statistics for entire policy instead of individual life. The rules of type of illness should be applied mutatis mutandis.

This simulation should be done for say 1 Lac times for each cell of age, SI and policy plan combination to get the risk premium.

Once we have computed risk premium the same should be loaded for commission, management expenses, profit, etc. to arrive at the final premium.

(8)

[50 Marks]
